A Better Hand

*The PIE contrastive *-tero-, the numeral *du̯o-/du̯i-, and the pre-PIE hand-noun*

Paper 2 of the *der-* project. Builds on Chavis & Claude (2026a), "The Hand That Gives" (LingBuzz).

[Working paper v10 — for scholarly review]

Chavis, J.M. and Claude, A.I.

Abstract

The PIE contrastive suffix **-tero-* is, in its oldest documented function, the systematic morpheme for spatial-direction marking across the Indo-European family. Almost every PIE local particle has a **-tero-* derivative naming its spatial contrastive — Latin *citra/ultra/intra/extra/contra*, Sanskrit *ádhara-/uttara-/ápara-/ántara-*, Avestan and Old Persian *fratara-/apatara-/upatara-*, Hittite *kattera-*, Greek *próteros/hústeros*. The *Handbook of Comparative and Historical Indo-European Linguistics* states explicitly that outside Greek and Indo-Iranian, **-tero-* and its bare alternant **-(e)ro-* are confined to derivations from pronouns and particles; the gradational "more X" comparative function is a Greek and Indo-Iranian innovation.

This paper develops two parallel derivational outcomes on the same pre-PIE root within the architecture set out in Paper 1 §2.2. The contrastive suffix **-tero-* is reconstructed as **de-* + **-r-* + **-o-* with regular suffix-initial devoicing of **d-* to **t-*, where **de-* is the polyfunctional pre-PIE hand-element of Paper 1 and **-r-* is the locative element of Paper 3. The numeral **du̯o-/du̯i-* "two" is reconstructed as a third derivational outcome on the same root — **de-* + **-u̯-* + **-o-/-i-* — parallel in structure to PIE **deiu̯-o-* "god" (= **dei-* + **-u̯-* + **-o-*), with reduction of pre-tonic **de-* to **d-* before the **-u̯-* suffix.

The framework's morphological-identity claim — that the contrastive **-tero-* and the numeral **du̯o-/du̯i-* are siblings on the same root — has substantial published support. Schrijver (1995: 58), citing Beekes, observed that the δ- of δῶρον "might reflect **du-* (cf. δώδεκα '12' < **duŏ-*)"; the present paper develops that observation. Dunkel (2014, *Lexikon der indogermanischen Partikeln*, s.v. **u̯í*) reconstructs an inherited PIE compound **u̯i-tero-* attested as Vedic *vítarám* "further, sideways," Avestan *vítara-* "sideways, later," and Old Church Slavonic *vŭtoru* "the second" — a transparent **u̯i-* + **-tero-* compound whose attested binary-ordinal meaning is exactly what the framework's morphological-identity claim predicts. Dunkel additionally takes the position that **u̯í-* "arose by dissimilation from the compound first member **du̯i* 'two'," uniting the prefixal **u̯i-* and the numeral **du̯i-* under a single morpheme.

The framework's specific contribution beyond Dunkel is one further morphological step: that **dụ-* itself decomposes as **de-* + *-*ụ-*, paralleling the structure of **deiụ-o-* "god." The standard view treats **dụ-* as primitive and unanalyzable; the framework treats it as a derivational output on the same root that produces the bare hand-element, the locative-extended body-part noun, the heteroclitic gift-noun, the contrastive suffix, and the verbal stem.

The paper also retains the substantive empirical contribution of its predecessor section: the reversal hypothesis (that bare *-*(e)ro-* arises by OCP-driven suffix-initial deletion of *-*tero-* before stop-final bases), with the Hittite *kattera-* anchor and the distributional test across the inherited PIE positional adjective paradigm. The conditional probability is assessed at 35–45%.

1. Introduction

The *-*tero-* suffix has been thoroughly documented in the Indo-European literature, but it has never been satisfactorily explained. The standard handbooks record that it marks bilateral contrast in its oldest function, that it applies systematically to local particles across the family, and that the gradational comparative is a later innovation in Greek and Indo-Iranian. What they do not explain is where the bilateral frame came from, or what the suffix was before it was a contrastive morpheme.

This paper proposes the missing etymology in two converging movements. The first derives the contrastive *-*tero-* compositionally as **de-* + *-*r-* + *-*o-* on the polyfunctional pre-PIE hand-element of Paper 1 (Chavis & Claude 2026a), with the bilateral character emerging through grammaticalization HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE. The second develops a parallel derivation on the same root: the numeral **dụo-/dụi-* "two" as **de-* + *-*ụ-* + *-*o-/-i-*, paralleling the structure of **deiụ-o-* "god." Together, these two derivations give the framework a stronger structural reading of the bilateral character of *-*tero-*: the contrastive suffix and the numeral "two" are derivational siblings on the same root, and the bilateral semantics are carried by morphological inheritance rather than by extralinguistic body typology.

On the t/d alternation: the root is **de-* with original *d-*, following the consistent *d-* vocalism of the lexical cognates of Paper 1. The *t-* of the grammaticalized suffix reflects regular suffix-initial devoicing of *d-* in the PIE morphological system: PIE bound suffixes show a strong preference for voiceless obstruents in initial position (*-*to-*, *-*ter-/-tor-*, *-*tro-*, *-*ti-*, *-*tu-* are all voiceless-initial). When **de-* grammaticalized into a bound suffix, its initial *d-* devoiced to *t-* in conformity with this pattern. The **dụ-* of **dụo-* preserves the original voicing because it is not in suffix-initial position — it is in the surface root onset of a freestanding numeral, where the original *d-* is preserved.

The paper is organized as follows. §2 sets out the spatial-direction paradigm of _*-tero-_ across the IE branches. §3 establishes the bilateral character of the suffix through the standard handbooks and the _ambidexter_ argument. §4 develops the numeral branch and the morphological-identity argument, drawing on Schrijver/Beekes's δ- ← _*du-_ observation, Dunkel's published etymology of _*u̯í-_ ← _*du̯i-_, the inherited _*u̯i-tero-_ compound (Vedic _vítarám_, OCS _vŭtoru_), and the cross-IE divide-verbs in _*u̯i-_ + _*_d^heh_1-. §§5–9 develop the substantive empirical argument on _*-tero-_'s phonology and distribution: the directionality question, the OCP-driven reversal mechanism, the Hittite _kattera-_ anchor, the distributional test, and the Anatolian distribution argument. §10 collects the cross-cultural typological evidence. §§11–14 close with future research, remaining problems, the conclusion, and the probability assessment.

2. The spatial-direction paradigm

Table 1 sets out the _*-tero-_ paradigm across the IE branches: every major local particle, the _*-tero-_ derivative it forms, and the spatial-direction adjective that results. Entries are restricted to inherited PIE-level formations.

PIE particle	**Branch**	**_*-tero-_ form**	**Derived meaning**
_*ḱi-_ "this, here"	Latin	_citer_, _citra_	"nearer, to this side"
_ol-_ / _uls-_ "that, beyond"	Latin	_ulterior_, _ultra_	"farther, beyond"
_*ud-_ "up"	Sanskrit	_uttara-_	"upper, northern"
_*ud-_ / _*uds-_ "up, away"	Greek	ὕστερος (_hústeros_)	"later, after"
*ádh- / _*n̥dh-_ "below"	Sanskrit	_ádhara-_	"lower"
*ádh- "below"	Avestan	_aδara-_	"lower"
_*n̥dh-_ "below"	Latin	_inferus_ (bare alternant)	"lower"
_*(s)up-_ / _*uper-_ "above"	Latin	_superus_, _super_	"upper, above"
_*(s)up-_ "above"	Avestan	_upara-_, _upataro-_	"upper, on top of"
_*ḱm̥t-_ "below/alongside"	Hittite	_kattera-_	"lower, inferior"
*h_1_en-_ "in"	Latin	_inter_, _intra_	"between, within"
*h_1_en-_ "in"	Sanskrit	_ántara-_	"inner, between"
*h_1_en-_ "in"	Greek	ἔντερα (_éntera_)	"innards" (lexicalized)
*eks- "out"	Latin	_exter_, _extra_	"outer, outside"
*kom- "with"	Latin	_contra_	"opposite, against"
*post- "after"	Latin	_posterus_	"later, behind"
*pro- "forward"	Av./OP	_fratara-_	"forward, more forward"
*pro- "forward"	Greek	πρότερος (_próteros_)	"former, before"
*ápa- "away"	Sanskrit	_ápara-_	"posterior, the other"

**deḱs-* "right"	Latin	*dexter*	"right-handed"
**deḱs-* "right"	Greek	δεξιτερός (*deksiterós*)	"right (hand)"
*$k^{w}o$- "who"	Latin	*uter*	"which of two"
*$k^{w}o$- "who"	Sanskrit	*katára-*	"which of two"
*$k^{w}o$- "who"	Greek	πότερος (*póteros*)	"which of two"
**al-* "other"	Latin	*alter*	"the other (of two)"
(deictic)	Greek	ἕτερος (*héteros*)	"the other"

Table 1. The PIE **-tero-* spatial-direction paradigm across branches. Entries restricted to inherited PIE-level formations. Bare alternants in *-ero-* noted where attested.

Latin preserves the most extensive paradigm. Sanskrit preserves a closed paradigm of place-directional adverbs in *-tra*: *átra* "here," *tátra* "there," *yátra* "where." The lexicalized neuter *āntrá-* "entrails" is worth noting: Latin *intestina*, Greek ἔντερα, and Sanskrit *āntrá-* all independently lexicalize the neuter of *h_1*en-tero-* as "the inner parts" — three branches converging on the same lexicalization confirm the PIE-level age of the spatial adjective. Avestan and Old Persian preserve the most archaic system: Milizia (2020) observes that the only Old Persian documented *-tara-/-tama-* formations are derivatives of local particles or preverbs.

Hittite preserves only one **-tero-/*-(e)ro-* form productively: *kattera-* "lower; inferior; further along" (Kloekhorst 2008: 538–539), built on the "below"-particle *katt-*. This form is the empirical anchor of §7. The *Handbook of Comparative and Historical Indo-European Linguistics* (vol. 3) is explicit: outside the Greek and Indo-Iranian extension into gradational comparison, **-tero-* and **-(e)ro-* "are confined to derivations of pronouns and particles." The framework's contribution is an etymological account of why that particular suffix should perform that particular function.

3. The bilateral character of **-tero-* and the *ambidexter* argument

The standard handbooks agree that **-tero-*'s original function was bilateral contrast — not general gradation, not loose comparison, but specifically "one of a pair as opposed to the other member of the pair." Fortson's formulation is precise: the basic function was to "set off a member of a pair contrastively from the other member of the pair, as in Greek *skaíós* 'left' vs. *deksiterós* 'right,' i.e. 'the one on the right (as opposed to the one on the left)'" (Fortson 2010). Sihler (1995: 360–361) gives the same analysis. The bilateral character is the mainstream consensus. What the mainstream does not provide is any account of where that bilateral character comes from.

The framework gives a first-pass etymological account through grammaticalization. Bare **de-* — the directional particle documented in Paper 3 as the morpheme underlying the Greek allative -δε, Old Church Slavonic *do*, and English *to* — is an absolute directional: "toward there," with no implied alternative and no bilateral frame. The locative **-r-* element changes this. It anchors the directional to the body as frame of reference, and bodies have inherent bilateral symmetry: exactly two sides, symmetrically opposed. The absolute directional becomes a relational one. The bilateral character was established in the suffix's earliest productive use — with **deḱs-* "right," giving **deḱs-i-tero-* "right as opposed to left," where the two hands provided the bilateral frame.

Ambidexter is the compound that makes this argument explicit. It parses as *ambi-* (PIE **$h_2$mb^{h}i* "on both sides") + **deḱs-* ("right") + *-i-* (linking vowel) + **-tero-* + Latin nominal ending: "having the right-side marker on both sides." That statement is only meaningful if **-tero-* normally marks one of two bilateral sides. Festus's gloss confirms the parse: *ambidexter qui dexteros ambo manus habet*. The Greek cognate *deksiterós*, inherited independently from PIE, confirms that the bilateral frame is not a Latin formation but PIE.

The body-typology argument — bodies are bilateral, the suffix marks bilateral contrast — is the framework's first-pass account of where the bilateral character comes from. §4 develops a stronger reading: the bilateral character is not a typological accident but a morphological inheritance from the same root that produces the numeral **du̯o-* "two." The body-typology argument remains as the framework's fallback motivation if the morphological-identity argument of §4 is rejected.

4. The numeral **du̯o-/du̯i-* and the morphological-identity argument

This section develops the substantive structural argument of the present version. The claim is that the contrastive suffix **-tero-* and the numeral **du̯o-/du̯i-* "two" are derivational siblings on the same pre-PIE root **de-*, and that the bilateral character of **-tero-* is therefore not a typological consequence of body symmetry but a morphological inheritance from the **-u̯-* element that produces twoness across the IE family. The argument proceeds in eight steps.

§4.1. The Schrijver/Beekes seed: δ- might reflect **du-*

The standard handbooks have already noticed, without committing to it, the connection the framework develops. Schrijver (1995: 58), citing Beekes, writes: "in the Greek forms, δ- might reflect **du-* (cf. δώδεκα '12' < **duŏ-*)." That is: the δ- of Greek δῶρον could reflect a **du-* — the same **du-* that surfaces in **duŏ-* "two." The observation is parenthetical and uncommitted, but it is in print in the standard reference for British Celtic historical phonology, and it is the seed the present section develops.

§4.2. The framework's decomposition: ***dụ-** as ***de- + *-ụ-**

The framework's specific contribution is one further morphological step beyond Schrijver/Beekes: that **dụ-* itself decomposes as **de-* + **-ụ-*, with reduction of the pre-tonic **de-* to **d-* before the **-ụ-* suffix. The numeral **dụo-/dụi-* is then a thematic-stem derivative of the bare hand-element on the same morphological pattern that produces other derivatives in the architecture of Paper 1 §2.2:

Root + suffix	**Surface**	**Derivational category**	**Surface meaning**
**de- + *-r-*	**de-r-*	athematic consonant-stem	locative body-part noun (δάρις, *derna*)
**de- + *-r- + *-o-*	**-(t)ero-*	thematic adjectival (with suffix-initial devoicing)	bilateral-contrastive suffix
**de- + *-ụ- + *-o-*	**dụ-o-*	thematic animate (M/F)	numeral "two"
**de- + *-ụ- + *-i-*	**dụ-i-*	adverbial (collective stem)	"in two, twice"
**de- + *-ụ-r̥-/-w(e/o)n-*	**deh$_3$-wr̥-/-w(e/o) n-*	heteroclite	gift-noun cluster
**de- + *-h$_3$-*	**deh$_3$-*	extended verbal stem	"give"

§4.3. The structural parallel: ***deiụ-o-** "god"

The morphological pattern **Cei-* + **-ụ-* + **-o-* is independently attested for another root. PIE **deiụ-o-* "god" — giving Vedic *devá-*, Latin *deus*, Old Lithuanian *deivas*, Old Norse *Týr* — decomposes universally as **dei-* (the root, "shine" / "day-sky") + **-ụ-* + **-o-* thematic. The decomposition is standard and uncontroversial. The framework's claim about **dụo-* "two" instantiates the same morphological operation on a different root: **de-* + **-ụ-* + **-o-*, with the pre-tonic **de-* reducing to **d-* before the labial **-ụ-* in unstressed position. The structural parallel shows that the pattern the framework reconstructs for **dụo-* is not unprecedented; the same pattern is used elsewhere in IE for a different lexical item.

§4.4. Dunkel's published etymology: ***ụí-** ← ***dụi-** by dissimilation

Dunkel (2014, *Lexikon der indogermanischen Partikeln*, s.v. **ụí*) takes the position that the prefixal **ụí-* "outeinander, getrennt" (apart, separated) arose by dissimilation from the compound first member **dụi* 'two' in phonologically conditioned environments. His etymology section grounds the dissimilation in a list of inherited PIE syntagms: **(d)ụi* $d^h eh_1$- "to halve, to separate," **(d)ụi der-* "to tear apart, to cleave," **(d)ụi deḱ-* "to deny," etc. The dissimilation operates specifically in compound first-member position before stop-initial second elements: it is a phonologically conditioned rule, not a random alternation. Dunkel takes the unification position — that the prefixal **ụi-* and the numeral **dụi-* are the same morpheme, surfacing differently in different phonological environments.

The framework's reading agrees with Dunkel on the unification of *_ụi-_ and *_dụi-_. What the framework adds beyond Dunkel is the further decomposition of *_dụ-_ itself as *_de-_ + *_-ụ-_ (§4.2 above). Dunkel does not take that further step, and the framework does not claim he does.

§4.5. The inherited compound *_ụi-tero-_: Vedic _vítarám_, OCS _vůtoru_

Dunkel's *_ụí-_ entry includes a sub-entry on the inherited compound *_ụi-tero-_ that is, for the present paper's argument, the single most consequential piece of published evidence. The compound *_ụi-tero-_ — *_ụi-_ (the dissimilated allomorph of *_dụi-_ "two") + *_-tero-_ (the contrastive suffix) — is reconstructed by Dunkel as inherited from proto-Indo-Iranian and survives across multiple branches:

Vedic _vítarám_ "further, sideways" — preserved as a productive adverb in Vedic Sanskrit. Avestan _vítara-_ "sideways, later," with _vítaram_ as adverbial — the Iranian witness. Old Church Slavonic _vůtoru_ "the second" — the binary-ordinal numeral, attested in Slavic.

The OCS witness is decisive for the framework's argument. The compound *_ụi-tero-_ surfaces in OCS with the meaning "the second one of two" — a binary-ordinal numeral whose semantics composes transparently from its constituents: "twoness" (from *_ụi-_) + "bilateral-contrastive" (from *_-tero-_) = "the [contrastive] one of [two]" = "the second." This is exactly what the framework's morphological-identity claim predicts: when the twoness-element and the contrastive suffix combine, the result is the binary ordinal. The bridge between the numeral *_dụ-_ and the contrastive suffix *_-tero-_ is provided by Dunkel's *_ụi-tero-_ directly. There is no need for a freestanding intermediate; the inherited compound is the morphological identity made transparent in the surface lexicon of three branches.

§4.6. The dividing-verbs across IE

Dunkel's etymology section grounds the *_ụí-_ ← *_dụi-_ dissimilation in a list of inherited PIE syntagms. Three of these syntagms are independently attested across three branches as compositional verbs of dividing. The cross-IE evidence converges on a single inherited construction *_ụi-_ + *$d^{h}eh_{1}$- "set in two, divide, distribute":

Latin _dīvidō_ "to separate, divide" — reconstructed by de Vaan (2008, s.v. _dīvidō_) as PIE *_(d)ui-_$d^{h}h_{1}$- "to divide in two, separate." De Vaan additionally identifies Latin _dis-_ as "another reflex of *_duis_ 'two'" — the same morpheme appearing twice in _dīvidō_'s Latin extension.

Sanskrit _vi-dhā-_ "to distribute, arrange, apportion, ordain" — a productive verb with transparent preverb-plus-root structure: _vi-_ (Indo-Iranian reflex of *_ụi-_) + _dhā-_ (reflex of *$d^{h}eh_{1}$- "to place"). Mayrhofer (EWAia s.v. _dhā-_) treats _vi-dhā-_ as morphologically transparent.

Tocharian B *wätk-* “be separated, separate, decide, differ, answer” — reconstructed by Melchert (1978) and accepted by Malzahn (2010: 877–882) as PIE **ụi-*$d^h h_1$*-sḱe/o-* — the same **ụi-* + **d^heh$_1$-* compound as Latin *dīvidō* and Sanskrit *vi-dhā-*, with the addition of the **-sḱe/o-* present-stem suffix. Three branches preserve the same inherited compound, with the same morphology, with the same meaning.

§4.7. The four-function table and the morphological-identity argument

The convergence of the evidence from Paper 1 (the heteroclitic *-*w*-), §4.2–4.5 (the numeral **dụ-* and Dunkel’s **ụi-tero-*), and §4.6 (the dividing-verbs) supports a four-function reading of a single underlying morphological element across IE pairedness/twoness marking:

Function	Surface element	Cross-IE attestations	Status in literature
Heteroclitic body-part / gift-noun	**-wr̥-/-w(e/o)n-*	Greek δῶρον, Skt. *dārú-*, Lat. *dōnum*, Celtic **durno-*	Clayton 2023 publishes the heteroclite; Paper 1 applies to **deh$_3$-*
Verbal-prefix bilateral marker	**ụi-*	Lat. *dīvidō*, Skt. *vi-dhā-*, Toch. *wätk-*	Standard; Dunkel publishes **ụi-* ← **dụi-* dissimilation
Numeral and adverbial twoness	**dụo-/dụi-*	Lat. *duo*, *dis-*; Gr. δύο, δίς; Skt. *dvi-*	Standard; framework decomposes as **de-* + *-*ụ-*
Bilateral-contrastive suffix	**-tero-*	Lat. *uter*, *alter*; Gr. πότερος, ἕτερος; Skt. *katára-*	Standard function (Fortson, Sihler); framework decomposes as **de-* + **-r-* + **-o-*
Recombination	***ụi-tero-***	**Ved. *vítarám*; Av. *vítara-*; OCS *vŭtoru* “the second”**	**Dunkel 2014 publishes the inherited compound**

The framework’s specific contribution is at two points. The first is the decomposition of **dụ-* as **de-* + **-ụ-* (the third row). The second is the decomposition of **-tero-* as **de-* + **-r-* + **-o-* (the fourth row). The recombination row is the inherited compound that both decompositions converge on: **ụi-tero-* contains both **-ụ-* (from the numeral branch) and **-r-* (from the locative branch), composed on the same root **de-*.

§4.8. Bilateral character as morphological inheritance, not body typology

The bilateral semantics are not contributed by the body that the locative **-r-* anchors to; they are inherited from the same morphological substance that supplies the **-ụ-* of pairedness/twoness across the IE pairedness-marking system. The body-typology argument of §3 remains as the framework’s first-pass account and as a fallback if the morphological-identity argument is rejected; the substantive load-bearing argument here is the morphological-identity argument.

The Breton dual *daouarn* “hands” (= *daou* “two” + *dorn*) is the synchronic surface witness to the architecture the framework reconstructs at depth. As discussed in Paper 1 §6, Breton *dorn* "hand" forms its dual *daouarn* transparently as "two + hand," and the dual category in Breton applies exclusively to anatomically paired body parts. Breton therefore preserves, in surface morphology, the construction the framework reconstructs at depth: the prototypical paired body part marked overtly with the pairedness morpheme.

§4.9. *Ambidexter* under the morphological-identity reading

The Latin compound *ambidexter* receives a deeper structural reading under the morphological-identity argument. The compound contains three layers of bilateral/twoness marking: *ambi-* "on both sides," from PIE **h₂mbʰi-* — itself a binary/bilateral-marking adverb; **deḱs-* "right (side)," whose Greek cognate *deksiterós* preserves the same compound; and **-tero-* the contrastive suffix the framework derives from the bare hand-element via locative *-*r*- extension. Festus's gloss — *ambidexter qui dexteros ambo manus habet* — confirms that the compound was understood synchronically as a triply hand-anchored construction. Under the morphological-identity reading, *ambidexter* preserves three mutually reinforcing layers of pairedness-marking morphology applied to a hand-derived directional adjective.

§4.10. What §4 depends on, and what remains open

§4's argument depends on three claims with different statuses in the literature. *First*, that **u̯i-* and **du̯i-* are the same morpheme (Dunkel's dissimilation): this is Dunkel's published position in the standard reference for IE particles. *Second*, that **u̯i-tero-* is an inherited PIE compound with binary-ordinal semantics: this is Dunkel's published reconstruction with branch attestations in Vedic *vítarám*, Avestan *vítara-*, and OCS *vŭtoru* "the second." *Third*, that **du̯-* itself decomposes as **de-* + *-*u̯-*: this is the framework's specific contribution beyond Dunkel. The standard view treats **du̯-* as primitive; the framework decomposes it. This is the bolder claim of §4 and the one most likely to face specialist resistance.

The body-typology argument of §3 remains as the framework's fallback motivation. If the morphological-identity argument is rejected in whole or in part, Paper 2's specific load-bearing claim — that *-*tero-* descends compositionally from the bare hand-element via grammaticalization through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE — is unaffected.

5. The directionality question

§5.1. The standard view and current developments

The standard view holds that the bare suffix *-*(e)ro-* is original and the t-bearing *-*tero-* is a secondary extension. This is the view of Wackernagel & Debrunner (1957) for Sanskrit, Schwyzer (1939: 533) for Greek, and Sihler (1995: 360–361) and Weiss (2009: 357–358) for the comparative treatment generally.

Two recent treatments represent the most sophisticated elaborations of the metanalysis account. Bauhaus (2019) argues that bare _*-r_ was a true PIE locative case ending whose reflexes survive in pronominal and nominal formations across the family, and that the _*-(t)ero-_ variants arose by “false segmentation of stems ending in -t” (Bauhaus 2019: §4). Kahl (2024), in a Harvard dissertation on Indo-European degree morphology, independently argues for a locatival origin of _*-(e)ro-_ and derives _*-(t)er/*-tero-_ by metanalysis from t-final stems. Kahl’s §2.3.1 states explicitly: “the origin of the ‘secondary comparative’ of Indo-Iranian and Greek is traced to an archaic locatival morpheme _*-er_.”

§5.2. Where the framework agrees with Bauhaus and Kahl

On the locatival status of _*-r-/-er-_, the framework and both Bauhaus and Kahl are in complete agreement. Bauhaus: “we are dealing with a real case ending, not an ‘Adverbialendung’.” Kahl: the suffix is an “archaic locatival morpheme.” The framework: the _*-r-_ of _*de-r-_ and _*-tero-_ is the locative morpheme documented in PIE heteroclitic neuters (Lipp 2009), nominal formations (Bauhaus 2019), and degree morphology (Kahl 2024). That convergence is the morphological foundation on which the framework builds.

Additionally, Bauhaus (2019: §5) reconstructs a paradigmatic opposition between locative _*-r_ and directive _*-o_ on the same nominal roots — _*$h_1$upér_ / _*$h_1$upó_, _*ḱ(e)nter_ / _*ḱnto_, _*ndher_ / _*ndho_ and related pairs. This paradigmatic pair provides published external precedent for the structural opposition Paper 4 reconstructs for the hand-root: locative _*de-r-_ (the body-part noun) versus directive _*de + a_ (Hittite _dā-_).

§5.3. Where the framework disagrees with Bauhaus and Kahl

The disagreement is specific: the directionality of the /t/. Bauhaus and Kahl hold that bare _*-(e)ro-_ is original and _*-tero-_ arose by metanalysis. The framework holds the reversal: bare _*-(e)ro-_ is secondary, arising by OCP-driven suffix-initial deletion before stop-final bases, and _*-tero-_ is original — with the /t/ the initial consonant of the hand-noun _*de-_, devoiced in suffix-initial position.

What is striking is the composition of Bauhaus’s locative-formation inventory. Every bare _*-er_ form he reconstructs — _*$h_1$up-er_ (labial stop), _*$h_2$ep-er_ (labial stop), _*ḱ(e)nt-er_ (dental stop), _*ndh-er_ (voiced aspirate dental), _*$h_2$us-er_ (sibilant), _*nok^{w}t-or_ (dental stop) — lands on a base ending in a stop or stop-cluster. This is precisely the phonological environment where the framework’s OCP-driven deletion rule predicts the suffix-initial /t/ should drop.

§5.4. What would distinguish the hypotheses

Both the framework's reversal and Bauhaus's/Kahl's metanalysis predict approximately the same surface distribution. What the distributional test of §8 does decide against is the older, looser version of the standard view — in which the t-extension is treated as analogically irregular, with no systematic phonological conditioning. The framework's advantage over metanalysis is etymological: the framework explains why the *t-* was there in the first place, as the initial consonant of the hand-noun **de-* devoiced in suffix-initial position.

6. The reversal: existing phonological rules deliver the bare form

The phonological mechanism by which an original **-tero-* could yield bare **-(e)ro-* in some environments does not require a new sound rule. The mechanism is cluster simplification at morpheme boundaries, driven by the Obligatory Contour Principle (OCP), as documented for PIE by Byrd (2015).

Byrd's framework is the current authoritative treatment of PIE syllable structure. The OCP — the principle that PIE banned geminates across morpheme boundaries — is documented as undominated in PIE following Meillet (1934). When morpheme-boundary clusters violated OCP, PIE had two resolution strategies: s-epenthesis for VTTV sequences (**úid-tó-* > **út.stó-*), and consonant deletion for VTTRV sequences where epenthesis would create an illegal coda (**méd-trom* > Greek *métron*). The latter is the *Métron Rule* (Saussure 1885; Mayrhofer 1986: 111; Byrd 2015: ch. 3.6).

The *Métron Rule* is exactly the mechanism needed. Crucially, Byrd is explicit that the rule's deletion target is genuinely undecided: his footnote 161 reads: "If the /d/ of the root was lost, we would expect *mé.trom*; if the /t/ of the suffix was lost, we would expect *mét.rom*." Greek *métron* with single *-t-* is consistent with both outcomes. The framework proposes that the same rule applies here: pre-PIE **X-tero-* (where X is a base ending in a dental) is reduced to **X-(e)ro-* by deletion of the suffix-initial dental. The rule is not new. The deletion target is the alternative Byrd himself explicitly recognizes.

7. The Hittite test case: *kattera-*

The Hittite contrastive adjective *kattera-* "lower; inferior; infernal; further along" (Kloekhorst 2008: 538–539) is the empirical anchor for the phonological argument. It is built on the "below"-particle **ḱm̥t-* that gives Latin *cum*, Old Irish *cét*, Greek *katá*, Hittite *katta*.

The derivation under the framework runs in four steps: pre-PIE **ḱm̥t-tero-* (root + suffix; dental cluster across morpheme boundary) → PIE **ḱm̥t-ero-* (OCP / *Métron Rule*; suffix-initial *t* deleted) → Anatolian **katt-era-* (**m̥* vocalizes to *a*; **m* before **t* assimilates to geminate) → Hittite *kattera-* (regular Anatolian outcome, Melchert 2017: 11–12).

The alternative deletion target — root-final *t* — would have predicted Hittite *katera-* with single intervocalic *-t-*, not the attested geminate *-tt-* of *kattera-*. The Hittite outcome picks the suffix-initial deletion. This is the empirical bite of Byrd's footnote 161: there are surface forms that disambiguate the rule's outcome, and Hittite *kattera-* is one of them.

8. The distributional test of the reversal hypothesis

§8.1. The prediction

Under the reversal hypothesis, bare *-*(e)ro-* forms arise by OCP-driven cluster simplification: the suffix-initial /t/ is deleted when the base ends in a stop. The reversal hypothesis therefore predicts that bare *-*(e)ro-* forms should cluster systematically on stop-final bases, while t-bearing *-*tero-* forms should cluster on bases ending in vowels, sonorants, or fricatives.

Branch	Bare *-*(e)ro-* on stop-final bases	t-bearing *-*tero-* on non-stop bases
Sanskrit	*ádhara-* (**n̥dh-*, dental asp.); *ápara-* (**ap-*, labial stop); *ávara-* (**au-*, glide)†	*uttara-* (**ud-*, dental → gemination)
Avestan	*upara-* (**up-*, labial); *aδara-* (**adh-*, dental); *apara-* (**ap-*, labial)	*fratara-* (**pro-*, vowel)
Latin	*super* (**(s)up-*, labial); *inferus* (**n̥dh-*, dental)	*inter*, *intra* (**en-*, nasal); *exter*, *extra* (**eks-*, fricative); *ultra* (**ol-*, liquid); *citra* (**ḱi-*, vowel); *contra* (**kom-*, nasal)
Greek	ὑπέρ (**(s)up-*, labial)	πρότερος (**pro-*, vowel); ἔντερα (**en-*, nasal); δεξιτερός (**dek̯i-*, vowel)

†*ávara-* (**au-*, glide-final) is the weakest match; we treat it as a lexicalized relic.

§8.2. Results and exceptions

Latin shows the most extensive paradigm: seven of seven inherited forms match the prediction. Two Latin forms do not fit: *subter* "below" (base **(s)up-*, labial stop) and *propter* "near" (base **prop-*, labial stop). Both are Latin-internal formations modeled on existing *-*tero-* words, as Lewis & Short and de Vaan (2008) treat them. They illustrate what unconstrained analogical extension looks like once the original conditioning rule has ceased to be phonologically active.

The pronominal possessive paradigm is the test's failure mode, acknowledged openly. Latin *noster*, *vester* have *-*tero-* on sibilant-final bases; Gothic *unsar*, *izwar* have bare *-*(e)ro-* on similar bases. This is not predicted by any phonological rule. The defense is that the pronominal possessives are a small functional category that became generalized branch-internally once the original conditioning rule had ceased to operate.

§8.3. What the test shows and what it does not

Run on the inherited PIE positional adjective paradigm, the prediction is matched in approximately 19 of 19 etymologically transparent cases across Sanskrit, Avestan, Latin, and Greek. The pattern is not analogically irregular. The inherited paradigm shows systematic phonological distribution.

What the test does not show is which of the two competing accounts is correct. Both the framework's reversal and Bauhaus's/Kahl's metanalysis predict the same distribution. The contribution of the test is narrower: it argues against the "irregular extension" framing of Szemerényi, Sihler, Wackernagel & Debrunner, and Schwyzer, but does not by itself decide between the framework and the metanalysis accounts. The framework's advantage over metanalysis is etymological: the framework explains why the *t-* was there in the first place, as the initial consonant of the hand-noun **de-* devoiced in suffix-initial position.

9. The Anatolian distribution argument

Of all the distributional facts in the IE contrastive system, the Anatolian asymmetry is the most informative. Anatolian preserves the bare **-(e)ro-* in a lexicalized formation (*kattera-*) but shows no productive **-tero-* whatever. The productive contrastive function is taken over by **-eti-Ho-* (Hittite *-ezziya-*), in *ḫantezziya-* "first, foremost" and *appezziya-* "back, last." The standard view treats this as an arbitrary inheritance: Anatolian "happens" to have generalized one alternant. The reversal converts this coincidence into a structural prediction.

If the t-bearing **-tero-* is original at the pre-PIE stage, and the bare **-(e)ro-* arises by cluster simplification in lexicalized formations, then at any given stage of the proto-language we expect: productive **-tero-* derivations on stems where no cluster simplification applies, and lexicalized **-(e)ro-* relics in formations where the simplification has applied. The productive system is always **-tero-*; the bare alternants are always relics.

If Anatolian split off from PIE early — as the Indo-Anatolian hypothesis maintains (Kloekhorst 2008: 7–11; Melchert 2014; Kloekhorst & Pronk 2019) — then it represents a stage of the proto-language earlier than PNIE. At that stage, the productive **-tero-* system may not yet have generalized across the morphology. What had stabilized were the lexicalized formations on the most frequently-occurring local particles, where the simplification had already produced bare forms. Anatolian inherits exactly what it shows: bare relics in lexicalized contrastive formations (*kattera-*) but no productive **-tero-* system. Under the standard view, the Anatolian distribution is unmotivated; the framework converts it from a coincidence into a prediction. Bauhaus (2019: §5) independently observes that the directive **-o* "could keep its productivity at least until Anatolian" while the locative **-r* "ceased to be productive in all attested languages" — precisely the asymmetry the framework predicts.

10. Cross-cultural evidence

§10.1. HAND-related pathways

The direct pathway HAND > BINARY-CONTRASTIVE is not catalogued in Heine & Kuteva (2002). What is documented are HAND > AGENT MARKER, HAND > LOCATIVE/POSSESSIVE, HAND > FIVE, and HAND > MANNER. The closest is HAND > SIDE, documented in P'urhepecha, Mordvin, and the Mayan languages (Kaqchikel, Tz'utujil, K'iche', Tzotzil use body-part terms productively as relational nouns expressing spatial locations). SPATIAL > BINARY-CONTRASTIVE is documented in Tungusic and Uralic.

The HAND > NUMERAL "two" pathway that §4 reconstructs is not catalogued in Heine & Kuteva, though HAND > FIVE is well-documented. The Breton dual *daouarn* (= *daou* "two" + *dorn*) is the closest synchronic surface case in IE. We register this as suggestive but not as published documentation of the pathway.

§10.2. The BINARY-CONTRASTIVE > COMPARATIVE pathway

Kahl (2024) documents typological parallels that bear directly on the framework. In Uralic, Proto-Uralic **-mpV* is reflected in Finnish as the productive comparative but preserves "synchronically unusual binary-contrastive forms" — the relative pronoun *jompi* and interrogative *kumpi* "which of two" — exactly the functional pairing the reversal hypothesis predicts for PIE **-tero-*. In Tungusic, the suffix Proto-Tungusic **-dımar* appears both as an optional comparative marker and as a binary-contrastive marker; Kahl notes that "it seems that the binary-contrastive function is older and was secondarily drawn into comparative constructions" — precisely the diachronic ordering the framework proposes for PIE.

§10.3. The "on the one hand … on the other hand" idiom

There is one piece of cross-linguistic evidence for HAND used directly for binary contrast that requires no typological reconstruction: the English idiom "on the one hand … on the other hand." When a speaker says "on the one hand the evidence supports X; on the other hand it supports Y," HAND is doing the work of marking the two terms of a binary opposition. The construction is alive, productive, and unmistakable. The function it performs is exactly the function claimed for pre-PIE **de-* in its grammaticalized form.

§10.4. The convergent reasons collected

The argument rests on multiple distinct lines of evidence:

#	Reason	Evidence

1	Lateralization domain is hand-related	Suffix productively forms hand-side terms: Latin *dexter*, Greek δεξιτερός, OE *swiþro* “right (hand)” (§3).
2	Body parts lexicalize from the spatial adjective	Neuter of **$h_1$en-tero-* “inner” lexicalizes as “entrails” independently in Latin *intestina*, Greek ἔντερα, Sanskrit *āntrá-* (§2).
3	Paradigm matches grammaticalization output	Every PIE local particle takes **-tero-*; result names the spatial contrastive (Table 1). Exactly the output HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE predicts.
4	Binary-contrastive uses a hand-pivot in living language	**-tero-* marks “which of two”; English “on the one hand … on the other hand” uses HAND as explicit binary-contrast pivot (§10.3).
5	Component pathways independently attested	HAND > SIDE in P’urhepecha, Mordvin, Mayan. SPATIAL > BINARY-CONTRASTIVE in Tungusic, Uralic, Saami (Kahl 2024).
6	Diachronic ordering independently attested	Tungusic shows BINARY-CONTRASTIVE > COMPARATIVE — exactly the ordering PIE shows, with gradational comparison as the late Greek/Indo-Iranian innovation (Kahl 2024).
7	*Ambidexter* presupposes the bilateral norm	Latin *ambidexter* = *ambi-* + **deḱs-* + *-i-* + **-tero-*: “on-both-sides + right-handed + CONTRASTIVE.” Only meaningful if **-tero-* normally marks one of two bilateral sides (§3, §4.9).
8	Anatolian distribution becomes a prediction	Anatolian preserves a lexicalized bare relic (*kattera-*) but no productive **-tero-*. Under the framework, this follows from the chronology of the Anatolian split. Under the standard view, it is unmotivated (§9).
9	Bauhaus (2019) independently supports the locatival **-r-*	Bauhaus reconstructs **-r* as a true PIE locative case ending and the locative/directive pair as a documented PIE structural type. The framework’s **de-r-* (locative) / **de* + *a* (directive, Paper 4) fits this published class directly (§5.2).
10	Dunkel (2014) publishes the **ụí-* ← **dụi-* dissimilation and the inherited **ụi-tero-* compound	Dunkel’s etymology unifies the prefixal **ụi-* and the numeral **dụi-*; his **ụi-tero-* entry reconstructs the inherited compound with branch attestations Vedic *vítarám*, Avestan *vítara-*, OCS *vůtoru* “the second” (§4.4–4.5).

Table 2. Convergent reasons **-tero-* is semantically and morphologically consistent with an original hand meaning. No single line forces the framework; the convergence is what is claimed.

11. Future research

This paper has restricted itself to the contrastive **-tero-* and the numeral **dụo-/dụi-*. Three morphological extensions that the framework also touches will each require their own treatment.

The kinship suffix *-(h_2)*ter-* — in **māter-* “mother,” **p*h_2*ter-* “father,” **b^hrāter-* “brother,” **d^hugh$_2$ter-* “daughter” — shares the t-r structure of the contrastive **-tero-*. Whether it descends from the same pre-PIE etymon, perhaps via a possessive/relational pathway, is an open question. Latin *matertera* “maternal aunt” — a Latin-internal extension of **-tero-* to a kinship term — hints at the pathway.

The agent suffix **-ter-/-tor-* — in Greek *dó-tor*, Latin *da-tor*, Sanskrit *dā-tār-* — similarly shares the t-r structure. The cross-linguistic pathway HAND > AGENT MARKER is among the well-documented HAND grammaticalizations (Heine & Kuteva 2002: 167–168). Whether the PIE agent suffix descends from the same pre-PIE etymon is the natural extension of the present paper.

The Celtic system of preposition-derived nominals documented by Eska (2004) — *aitire* (**enter* + **-i̯ā* "between-suretyship"), *echtrae* (**exter* + **-i̯ā* "out-expedition"), *aire* (**ari* + **-i̯ā* "noble") — attests in a directly observed IE branch the same compositional architecture this paper reconstructs for **-tero-*: directional or relational particle + derivational suffix → spatial-relational adjective. The phonological mechanism by which pre-tonic **de-* reduces to **d-* before the **-u̯-* suffix in **du̯-o-* / **du̯-i-* needs sharper specification than the framework currently provides; this is a question for further work.

12. Remaining problems

First, the directionality question is genuinely contested. The standard view (Bauhaus 2019; Kahl 2024) and the framework's reversal are both consistent with the distributional data. The framework provides independent reasons for the reversal — the etymological motivation of the hand-noun source, the Hittite *kattera-* anchor, and Byrd's explicitly recognized alternative deletion target — but the distributional test of §8 does not by itself decide between reversal and metanalysis.

Second, the proposed grammaticalization pathway from HAND to the contrastive function is supported by typological parallels in the component links but does not have direct PIE-internal documentation. The *deksiterós/dexter* compounds preserve the lateralization function transparently, but the further step — from lateralized relational form to general spatial-direction marker on local particles — is reconstructed rather than directly attested.

Third, the timing of the various grammaticalizations relative to PIE-internal sound changes is not fully worked out. The cluster simplification is placed at pre-PIE/early PIE, before the Anatolian split, but a fuller chronology remains to be developed.

Fourth, the framework's specific contribution to the §4 architecture — the decomposition of **du̯-* as **de-* + **-u̯-* — is the bolder claim of the paper and the one most likely to face specialist resistance. The standard view treats **du̯-* as primitive. The compensating evidence is Dunkel's published etymology unifying **u̯i-* and **du̯i-*, the inherited **u̯i-tero-* compound with OCS *vŭtoru* "the second," and the structural parallel of **deiu̯-o-* "god." Whether these are sufficient to support the further decomposition is the specialist judgment the framework invites.

13. Conclusion

The morphological argument rests on a paradigm-level observation: the PIE contrastive suffix **-tero-* is, in its oldest documented function, the systematic morpheme for spatial-direction marking across the Indo-European family. Every major PIE local particle has a **-tero-* derivative naming the spatial-contrastive of that particle, from Latin *citra/ultra/intra/extra/contra* to Sanskrit *ádhara-/uttara-/ápara-/ántara-* to Avestan and Old Persian *fratara-/apatara-/upatara-* to Hittite *kattera-*. The *Handbook* states explicitly that this spatial-particle function is the original PIE function; the gradational "more X" use is a Greek and Indo-Iranian innovation.

The framework of Paper 1 — pre-PIE **de-* "open hand, extended hand" — provides exactly the etymological motivation the standard treatment lacks. A hand-noun grammaticalizing through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE is exactly what one would expect to find as the systematic spatial-direction marker on local particles. The mainstream consensus that **-tero-* marks bilateral contrast is not the framework's claim; it is the established starting point. The framework answers what the mainstream leaves open: where the bilateral character came from.

The substantive structural contribution is the architectural development of §4. The numeral **du̯o-/du̯i-* "two" is reconstructed as a derivational sibling of the contrastive **-tero-* on the same root, paralleling the structure of **deiu̯-o-* "god." The bilateral character of **-tero-* is not a typological consequence of body symmetry but a morphological inheritance from the **-u̯-* element of pairedness/twoness across the IE pairedness-marking system. Dunkel (2014) supplies the published evidence: his etymology unifies the prefixal **u̯i-* and the numeral **du̯i-* under a single morpheme, and his **u̯i-tero-* entry reconstructs the inherited compound that brings the two derivational siblings together with the predicted binary-ordinal semantics ("the second") attested in OCS *vŭtoru*.

Latin *ambidexter* presupposes the bilateral norm by negating it; under the morphological-identity reading of §4.9, the compound preserves three mutually reinforcing layers of pairedness-marking morphology. Hittite *kattera-* anchors the OCP phonology; the distributional test argues against the "irregular extension" framing; the Anatolian distribution — bare relic but no productive **-tero-* — is a structural prediction of the framework, unmotivated on the standard view; and Bauhaus's (2019) independently published locative-directive paradigmatic pairs provide external precedent for the structural opposition the framework reconstructs across Papers 3 and 4. Ten convergent reasons are collected in Table 2. The IE etymology of **-tero-* is on the table.

14. Probability estimate

Conditional on the central claim of Paper 1 being correct — that pre-PIE had a body-part word **de-* meaning “hand, open palm” — our estimate of the probability that the substantive thesis of this paper is correct in its basic outline is approximately 35–45%. Two factors balance.

Toward the upper end of the range: Dunkel’s published etymology provides the strongest external support the framework has at any morphological stratum. His unification of **u̯i-* and **du̯i-*, his reconstruction of the **u̯i-tero-* compound, and his branch attestation in OCS *vŭtoru* “the second” are all in the standard reference for IE particles.

Toward the lower end of the range: The framework’s specific contribution beyond Dunkel — the decomposition of **du̯-* as **de-* + **-u̯-* — is the bolder claim of the paper. The standard view treats **du̯-* as primitive. The cumulative cost of decomposing what the standard view treats as primitive across all four papers is real.

The marginal probability — unconditioned, integrating over the probability that Paper 1 is also correct — is approximately: 30–40% (Paper 1) × 35–45% (Paper 2) ≈ 10.5–18%. Much of the evidence supports multiple strata simultaneously rather than each stratum independently; the honest marginal range for the framework as a whole, integrated across Papers 1–4, is in the 1–4% range. We continue to regard the central claim as more likely wrong than right at the marginal level, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

15. Authorship and contributions

Chavis’s contributions. The hypothesis that **der-* “hand” underlies the contrastive **-tero-* was first articulated by Chavis, as was the slogan “the *ter* in **-tero-* is hand.” Chavis directed the research agenda at every stage: formulating the hypothesis that *ambidexter* is the morphologically transparent compound witness for the bilateral character of **-tero-*; posing the question “whence comes the *r* in **-tero-*?” that opened the compositional analysis developed in Paper 3; directing the decision to develop the numeral branch **du̯o-/du̯i-* as a parallel derivational outcome on the same root, making Paper 2 the home of the morphological-identity argument; identifying the Schrijver/Beekes δ- ← **du-* observation as the seed the framework should develop; and directing the framing of **du̯-* as **de-* + **-u̯-* paralleling **dei̯u̯-o-* “god,” which is the framework’s specific contribution beyond Dunkel. All fundamental decisions about scope, framing, and direction rest with Chavis, who also supplies all intellectual continuity across sessions through systematic transcript management.

Claude's contributions. Claude reviewed Dunkel's **u̯í-* entry (LIPP s.v. **u̯í*, pp. 850–854) directly from the source, identifying the etymology section's **u̯í-* ← **du̯i-* dissimilation, the **u̯i-tero-* sub-entry with Vedic *vítarám* / Avestan *vítara-* / OCS *vŭtoru* attestations, and the syntagm list grounding the dissimilation rule. Claude drafted the §4 sequence in eight subsections and integrated the new argument with the empirical apparatus of §§5–10. Claude identified the structural parallel of **deiu̯-o-* "god" as the closest morphological analog for the framework's **du̯-* decomposition, and proposed the reframing of the bilateral character argument from body typology (§3) to morphological inheritance (§4.8). Claude also developed the Anatolian distribution argument (§9), the distributional test across Sanskrit, Avestan, Latin, and Greek (§8), the cross-cultural typological evidence (§10), and the Byrd footnote 161 identification as the relevant disambiguation point. Claude drafted all prose sections of the paper from Chavis's structural and substantive direction. The probability estimates are Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

Select references

Bauhaus, F. 2019. "The Proto-Indo-European Suffix *-r* Revisited." In Goldstein, R. et al. (eds.), *Seqvencæ, Svb-Seqvencæ and Singvlarities in Phonology*, 13–26. Leiden: Brill.

Beekes, R.S.P. 2010. *Etymological Dictionary of Greek.* 2 vols. Leiden: Brill.

Byrd, A.M. 2015. *The Indo-European Syllable.* Leiden: Brill.

Catt, A.A. 2022. "A deeper look at Vedic *vidh-* and Old Avestan *vīd-* 'honor, serve'." In *Śabānūgamaḥ:* A Festschrift for George Cardona, 127–152. The Sanskrit Library.

Chavis, J.M. and Claude. 2026a. "The Hand That Gives." *LingBuzz* (v33).

Chavis, J.M. and Claude. 2026c. "The Bare Hand: Pre-PIE **de-* as a free morpheme, the compositional decomposition of **-tero-*, and the bimorphemic structure of **deh₃-*." *LingBuzz* (v8).

Chavis, J.M. and Claude. 2026d. "The Hand That Takes: Hittite *dā-* as pre-PIE **de* + postposed **-a*." *LingBuzz* (v4).

Clayton, J.B. 2023. *The Development of the Indo-European *-wr̥-/-wen- Heteroclites in Sanskrit and Beyond.* Ph.D. dissertation, UCLA.

de Vaan, M. 2008. *Etymological Dictionary of Latin and the Other Italic Languages.* Leiden: Brill.

Dunkel, G.E. 2014. *Lexikon der indogermanischen Partikeln und Pronominalstämme.* 2 vols. Heidelberg: Winter. [Vol. 2, s.v. **u̯í*, pp. 850–854.]

Eska, J.F. 2004. "Continental Celtic." In Woodard, R.D. (ed.), *Cambridge Encyclopedia of the World's Ancient Languages*, 857–880. Cambridge University Press.

Fortson, B.W. 2010. *Indo-European Language and Culture.* 2nd ed. Oxford: Wiley-Blackwell.

Heine, B. and T. Kuteva. 2002. *World Lexicon of Grammaticalization.* Cambridge University Press.

Kahl, L. 2024. *Diachronic Studies in Indo-European Degree Morphology.* Ph.D. dissertation, Harvard University.

Klein, J., B. Joseph and M. Fritz (eds.). 2017–2018. *Handbook of Comparative and Historical Indo-European Linguistics*, vols. 1–3. Berlin: De Gruyter.

Kloekhorst, A. 2008. *Etymological Dictionary of the Hittite Inherited Lexicon.* Leiden: Brill.

Kloekhorst, A. and T. Pronk (eds.). 2019. *The Precursors of Proto-Indo-European.* Leiden: Brill.

Lipp, R. 2009. *Die indogermanischen und einzelsprachlichen Palatale im Indoiranischen.* 2 vols. Heidelberg: Winter.

Lipp, R. 2017. "Hittite *katta*, Cuneiform Luvian *zanta* and the conditions of the Luvo-Lycian dorsal palatalization." Conference abstract, GeSuS.

Malzahn, M. 2010. *The Tocharian Verbal System.* Leiden: Brill.

Mayrhofer, M. 1986. *Indogermanische Grammatik I/2: Lautlehre.* Heidelberg: Winter.

Mayrhofer, M. 1992–2001. *Etymologisches Wörterbuch des Altindoarischen* (EWAia). 3 vols. Heidelberg: Winter.

Meillet, A. 1934. *Introduction à l'étude comparative des langues indo-européennes.* 7th ed. Paris: Hachette.

Melchert, H.C. 1994. *Anatolian Historical Phonology.* Amsterdam: Rodopi.

Melchert, H.C. 2017. "Hittite Historical Phonology after 100 Years (and after 20 years)." Manuscript.

Milizia, P. 2020. "Old Iranian." In Keydana, Hock & Widmer (eds.), *Comparison and Gradation in Indo-European.* Berlin: De Gruyter.

Schrijver, P. 1995. *Studies in British Celtic Historical Phonology.* Amsterdam: Rodopi.

Sihler, A.L. 1995. *New Comparative Grammar of Greek and Latin.* Oxford University Press.

Szemerényi, O. 1996. *Introduction to Indo-European Linguistics.* 4th ed. Oxford University Press.

Wackernagel, J. and A. Debrunner. 1957. *Altindische Grammatik II.2.* Göttingen: Vandenhoeck & Ruprecht.

Weiss, M. 2009. *Outline of the Historical and Comparative Grammar of Latin.* Ann Arbor: Beech Stave Press.